

Chasing the Wind: A GIS-Based Assessment of Wind Farm Siting Potential in the Penrith Region

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Introduction

Meeting the UK's clean power 2030 target demands fully decarbonising electricity production, placing onshore wind as the cornerstone of the transition away from fossil fuels (DESNZ, 2025). Selecting appropriate wind turbine sites that provide maximum energy potential requires simultaneous evaluation of multiple spatial criteria: wind resource, terrain, proximity to urbanisation, environmental constraints, and visual impact on communities. This geographical puzzle is well suited to Geographic Information System (GIS) analysis. GIS provides a powerful framework for multi-criteria spatial analysis, enabling the simultaneous evaluation of environmental, physical and social factors efficiently and reproducibly (Longley et al., 2015). This study uses a GIS approach to assess wind farm siting potential across study area a62 within the Penrith region, Cumbria. Three top locations were identified based on the potential power a wind turbine might generate using wind speed downscaling, suitability screening, energy ranking and viewshed analysis.

Methods

Study Area

Study area a62 is a rectangular sub-region of the wider Penrith area in Cumbria, northern England. The Ordnance Survey (OS) grid references defining the a62 study area are (334349, 509087) for the south-west corner and (386094, 586668) for the north-east corner. All spatial analysis was performed to the full Penrith extent and clipped afterwards to a62 to avoid unwanted edge effects.

Datasets

Datasets included a 30 m SRTM DEM; 5 km Met Office mean April wind speed data (1981–2010); a 25 m CEH Land Cover Map; OS road network and settlement point data; National Park boundaries; LSOA boundaries, and 2011 Census population data. All layers were projected to British National Grid (EPSG: 27700).

Analysis:

All spatial analysis was performed in ArcGIS Pro (3.6.0) (see Appendix 1 for full step-by-step details).

Wind speed data at 5km resolution was downscaled to 30m using a power law wind profile and the zonal mean elevation of each 5km zone was calculated. The ratio between each cell's elevation plus turbine hub height (80m) against the zonal mean

elevation plus measurement height (10m), was raised to the power of 0.143 (atmospheric stability coefficient), and multiplied by wind speed converted from knots to m/s. This produced hub-height wind speed estimates at 30m resolution.

Power generation estimation was calculated using Siemens turbine power curve. For wind speeds between 3 and 13m s⁻¹, power output (kW) was calculated as (230 x wind speed) – 690. Cells below 3m s⁻¹ were assigned 0. Outputs were multiplied by 720 hours to create April monthly energy production per turbine.

Six suitability constraints were applied: elevation below 300m, gradient below 10 degrees, within 3km of a road, more than 2km from urban/suburban land cover, more than 500m from named settlements, and outside National Park boundaries. Euclidean distance surfaces were combined with terrain and landcover rasters in a single conditional overlay to create a binary suitability mask, applied to the energy raster to generate only suitable cells.

Top-generating sites were discovered by classifying the suitable energy raster into 200 equal-area slices, and the top 0.5% selected and converted into polygons. These candidates were ranked by mean energy output using zonal statistics, and the top three sites were selected.

Viewshed analysis with a 20km radius identified which areas have line-of-sight to the three sites. Zonal statistics quantified how much of each LSOA was visible, which was joined to 2011 Census data to estimate number of affected individuals.

Results

Wind Potential Energy

Across study area a62, wind energy generation potential ranged from 0 to 1,034MWh per turbine per month in April, based on mean wind speeds recorded between 1981 and 2010. Topography drives the greatest influence, seen where elevated ridges around the North Pennine (SE) and Lake District (SW) provide the greatest energy potential (Figure 1). Lower energy potential areas were recorded in central and northwest lowland areas, specifically around urban areas such as Carlisle, likely due to a mix of poor elevation and reduced wind speed from urbanisation. The blocky pattern visible in Figure 1 reflects the 5km resolution of the source wind data, which remains a limitation even after downscaling.

Wind Energy Generation Potential Across Study Area a62, April

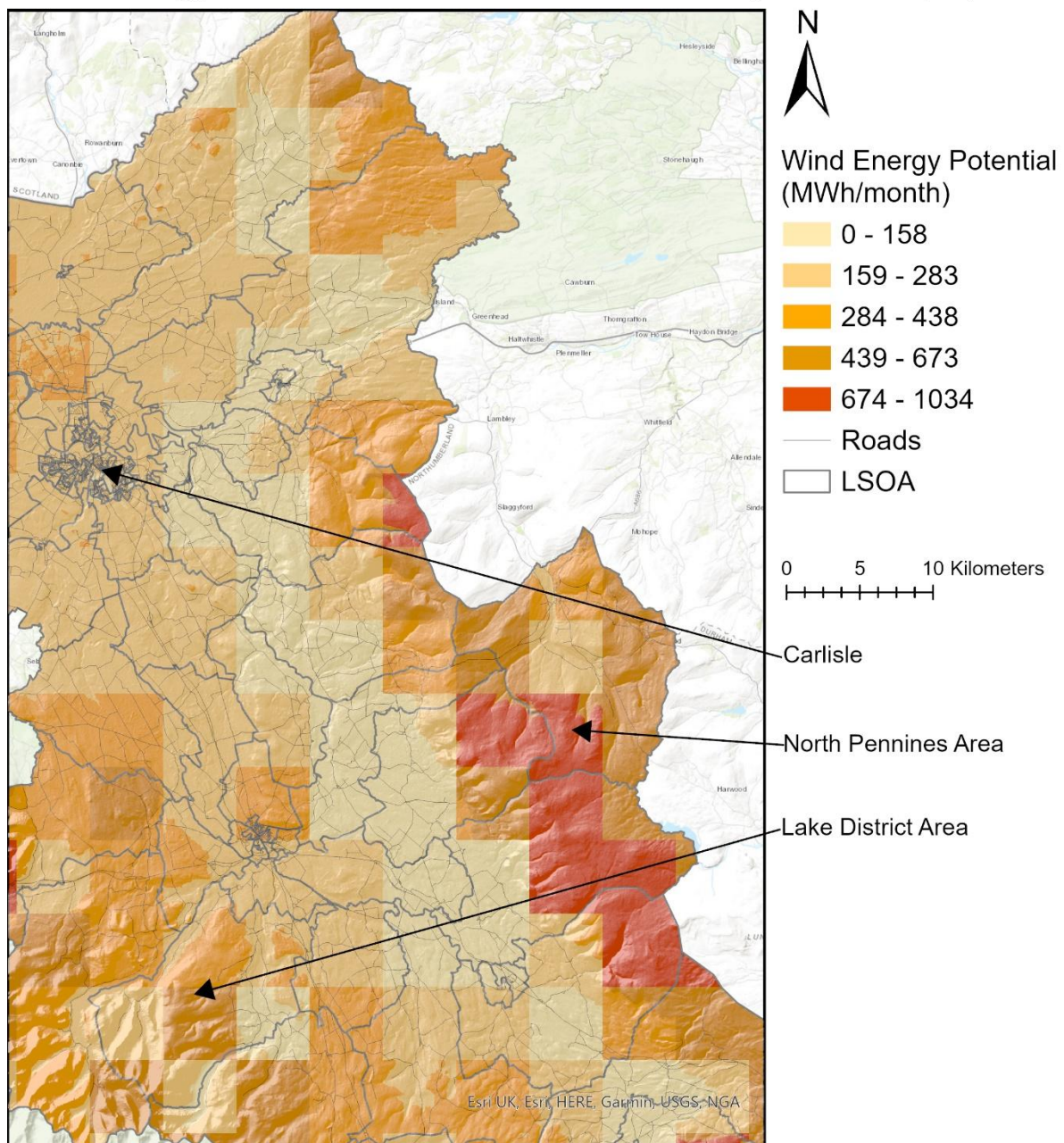


Figure 1: Estimated wind energy generation potential across study area a62, based on mean April wind speeds (1981–2010, Met Office). The original wind speed data at 5km was downscaled to 30m resolution using a power law wind profile. This was adjusted for each cells elevation relative to the mean elevation of its 5km zone, at a turbine hub height of 80m. Siemens turbine power curve ($\text{Power} = 230 \times \text{wind speed} - 690 \text{ kW}$) was applied to wind speeds between 3-13 m/s to calculate energy output, which was multiplied by 720 hours to obtain monthly energy production. Values range from 0-1034 MWh per turbine per month across a62. LSOA boundaries, hillshade, labels, and road networks are shown for geographic reference.

Suitability Analysis and Site Selection

The six suitability constraints (see Methods) substantially reduced area availability for turbine placement, with National Park exclusion causing the greatest loss of potential terrain (~19.8%). Most remaining suitable areas concentrate along the northern edge of the study area, as the high-gradient terrain of the North Pennines is largely unsuitable.

The top 0.5% of suitable cells produced nine candidate polygons, of which the top three mean energy outputs were selected (Table 1). Sites 1 and 2 are located near ridge lines in the north, just outside the Northumberland National Park (Figure 2), whilst Site 3 falls within the North Pennines AONB (Figure 2). Although site 3 is the smallest candidate area at only 900m² (one 30m pixel) it scores the highest mean energy output at 523 MWh per turbine per month, indicating a single suitable concentrated peak wind resource. Sites 1 and 2 fall on coniferous woodland and Site 3 on neutral grassland, each holding different land use and ecological implications.

Site	Area (m ²)	Land Cover	Mean Energy Output (MWh/month)
Site 1	9,900	Coniferous woodland	501
Site 2	2,700	Coniferous woodland	502
Site 3	900	Neutral grassland	523

Table 1: The three top-ranked candidate turbine sites within study area a62, showing their area, land cover, and estimated mean energy output for April.

Suitable Areas and Top Three Candidate Turbine Sites Within a62

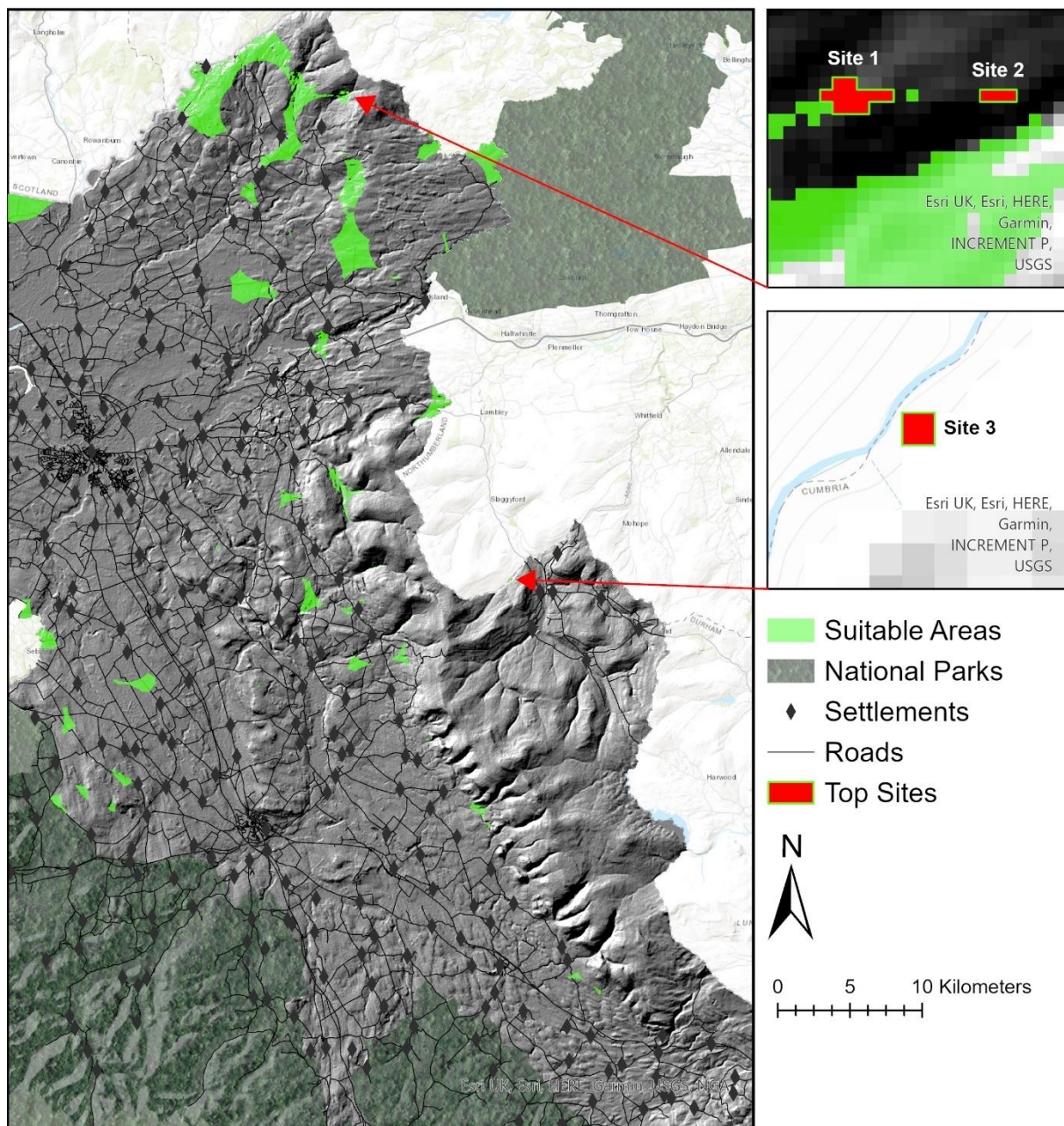


Figure 2: Suitable areas for wind turbine placement and the three top-ranked candidate sites within study area a62. Suitability was determined by excluding areas on gradients exceeding 10 degrees, above 300m elevation, within 2km of urban or suburban land cover (CEH classes 22-23), within National Park boundaries, within 500m of settlements, and more than 3km from roads. Suitable areas are displayed in bright green against a hill shade terrain model developed from the 30m DEM. The three top sites were selected for as the highest mean energy-producing polygons from the top 0.5% of suitable cells. Site 1 covers 9900m², Site 2 covers 2700m², and Site 3 covers 900m². Upper inset highlights Sites 1 and 2; the lower inset highlights Site 3.

Visibility and Population Impact

Viewshed analysis identified that the top three sites have a limited visual footprint across study area a62. Of the 104 LSOAs analysed, only 10 (9.6%) had any line-of-sight to the candidate sites, all falling within Carlisle or Eden areas with zero visibility across the south/southwest. Of the total population of 160,088 (Census 2011), only ~1,225 residents (0.8%) have any visibility to the proposed sites (Table 2), a figure attributable to the urban and settlement proximity constraints. This is a successful outcome, showing the candidate sites would have minimal visual impact.

The most visually exposed LSOA was Carlisle 002C, where 47.5% of the area has line-of-sight to the proposed sites, with roughly 782 residents affected at a density of 14.7 persons per hectare. In contrast, Carlisle 001D recorded the greatest raw visibility count (SUM = 32,637 cells) but due to its density of just 0.1 persons per hectare, only around 245 people are affected. This highlights the importance of proportional over absolute visibility metrics, as large rural LSOAs accumulate higher raw visibility counts due to size, whereas smaller denser LSOAs display more concentrated community exposure (Table 2).

Visibility Category	No. LSOAs	Total LSOA Population	Est. Residents with Visibility	% of Category Population Affected
None	94	142,697	0	0%
Low	3	4,452	18	0.4%
Medium	3	5,356	805	15.0%
High	2	3,648	95	2.6%
Very High	2	3,935	308	7.8%
Total	104	160,088	~1,225	0.8%

Table 2: How visible the three proposed turbine sites are across LSOAs in study area a62, broken down by visibility category. The estimated resident's column shows how many people within each category are thought to have line-of-sight to the sites, based on 2011 Census population data.

Visual Exposure to Proposed Turbine Sites Across Study Area a62

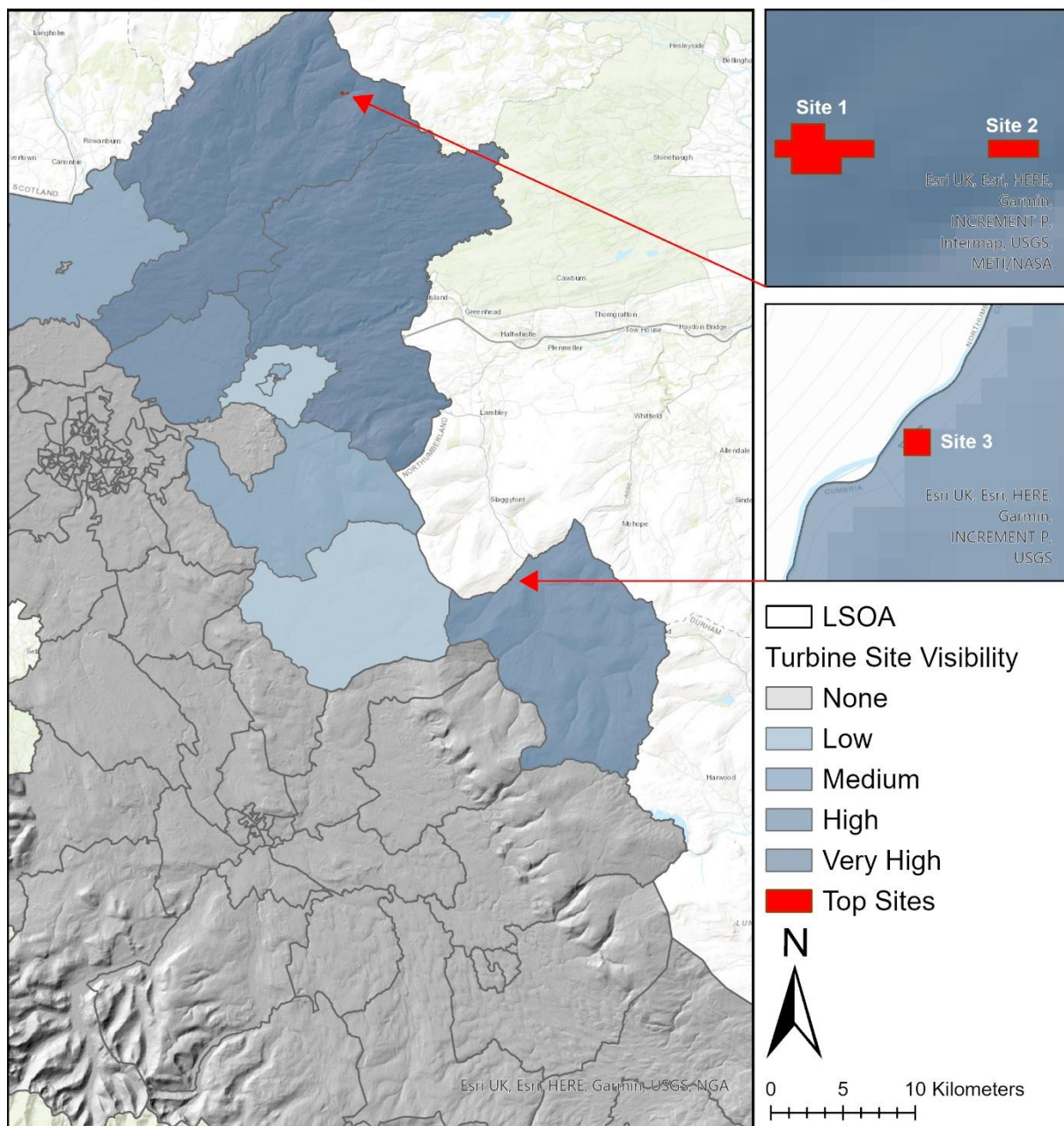


Figure 3: Visual exposure to the three proposed candidate turbine sites across LSOAs within study area a62. The boundaries of the three candidate site polygons were put under a viewshed analysis to determine visibility. The 30m DEM was used at 20km maximum viewing distance at 80m turbine height. Five qualitative categories were classified using the number of visible 90m grid cells within each LSOA. No visibility (0 cells), Low (1–197 cells), Medium (198–1,175 cells), High (1,176–2,753 cells), and Very High (2,754–32,637 cells). Darker shading indicates greater visual exposure to the proposed top site locations. Insets and red polygons highlight the three candidate sites as in Figure 2.

Discussion

The spatial distribution seen in Figure 1 reflects the major interaction between topography and wind resource, with the highest outputs observed along elevated ridges of the North Pennines AONB. This pattern is seen in existing wind farms across northern England, where development clusters on south-westerly upland terrain (RenewableUK, 2025). The block pattern in Figure 1 is a key limitation to the 5km Met Office wind speed data, as after downscaling to 30m, wind speed variation within the 5km zone is driven by terrain adjustment rather than an observed spatial variation.

The applied suitability constraints greatly reduced the available area for turbine placement. The Northumberland National Park in the northeast and the Lake District and Yorkshire dales in the south remove large portions of moderate energy potential. The NPPF protects National Parks from development (MHCLG, 2024), directing potential wind development to the upland fringe where Sites 1 and 2 are located.

The top three sites all produced similar mean energy outputs, ranging from 501 to 523 MWh per turbine per month, with Site 3 providing the greatest output despite being the smallest at 900m², just one 30m pixel. In practice, commercial onshore wind turbine sites typically cover tens of hectares to make them meaningful (Denholm 2009), far greater than any of the top three sites. These fractured sites are the consequence of the 0.5% threshold at 30m resolution, resulting in only peak energy cells rather than workable development areas. A more operationally realistic analysis could both relax the 0.5% threshold and apply a gap tolerance to aggregate proximate high-performing cells into coherent functional zones.

Sites 1 and 2 are located on coniferous woodland which is often managed commercial land use for felling cycles. These sites may present fewer ecological objections than Site 3, which is located on neutral grassland, a UK Biodiversity Action Plan Priority Habitat (JNCC, 2011). Site 3 also falls on the North Pennines AONB, which despite not being a national park, still faces serious planning constraints which makes it less favourable, particularly given the marginal energy gain.

The 2011 Census population data present a methodological limitation, at 15 years old it creates demographic uncertainty into the visibility impact estimates. Another key limitation is the wind speed data at 5km resolution, as cells within the same zone receive identical base wind speeds despite being at different topographic positions. Future work should aim to incorporate higher-resolution wind data, monthly energy potentials, current census figures and minimum site sizes to produce more applicable turbine sites.

References:

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Appendix:

This appendix gives step by step instruction using the tools in ArcGIS Pro (version 3.6.0) to complete the study reported. All datasets were made available in a folder on the computer, data-raw, which was split into 4 sub folders : a62, wind-speed, national-parks, penrith. At each stage used in the analysis, new map data was stored in a single file geodatabase called exam.gdb. Unless stated otherwise, all Environment settings (cell size and extent) should be matched to dem.tif. Analysis was performed across the entire Penrith area before being clipped into study area a62.

Start of Appendix

1. Project Setup and Loading Wind Speed Data

- Open ArcGIS Pro, create new project with file geodatabase (exam.gdb).
- Add Data: dem.tif
- Data Management Tools | Raster | Raster Datasets | Copy Raster
 - Input: MeanWindSpeed_Apr_1981-2010_LTA.asc
 - Output: windspeed_apr
 - Environments → Output Coordinate System: dem.tif

2. Downscaling Wind Speed to 30 m Resolution

- Conversion Tools | From Raster | Raster to Point

- Input: windspeed_apr | Output: points5km
- Conversion Tools | To Raster | Feature to Raster
 - Input: points5km | Value field: pointid | Cell size: 5000 | Output: zones5km
- Spatial Analyst Tools | Zonal | Zonal Statistics
 - Zone raster: zones5km | Value raster: dem.tif | Statistic: MEAN | Output: demmean
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{Power}(((\text{"dem.tif"} + 80) / (\text{"demmean"} + 10)), 0.143) * (\text{"windspeed_apr"} * 0.51444)$
 - Output: windspeedms

3. Estimating Power and Energy Generation

- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{Con}(\text{"windspeedms"} > 3, (230 * \text{"windspeedms"} - 690), 0)$
 - Output: powerKW
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{"powerKW"} * 720$
 - Output: energy_apr

4. Screening Out Unsuitable Areas

- Spatial Analyst Tools | Surface | Slope
 - Input: dem.tif | Output measurement: DEGREE | Output: gradient
- Spatial Analyst Tools | Distance | Legacy | Euclidean Distance
 - Input: roads.shp | Output: roadsdist
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{Con}(\text{"landcov.tif"} == 22) | (\text{"landcov.tif"} == 23), 1)$
 - Output: urban
- Spatial Analyst Tools | Distance | Legacy | Euclidean Distance
 - Input: urban | Output: urbdist
- Spatial Analyst Tools | Distance | Legacy | Euclidean Distance
 - Input: settlements.shp | Output: settldist
- Conversion Tools | To Raster | Feature to Raster
 - Input: national_parks.shp | Value Field : FID_1 | Output: natpark_r
- Spatial Analyst Tools | Map Algebra | Raster Calculator — combine all constraints:
 - $\text{Con}((\text{"dem.tif"} < 300) \& (\text{"gradient"} < 10) \& (\text{"roadsdist"} \leq 3000) \& (\text{"urbdist"} > 2000) \& (\text{"settdist"} > 500) \& \text{IsNull}(\text{"natpark_r"}), 1)$
 - Output: suitablemask
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{Con}(\text{"suitablemask"} == 1, \text{"energy_apr"})$
 - Output: energy_suitable

5. Identifying Top Candidate Sites

- Spatial Analyst Tools | Reclass | Slice

- Input: energy_suitable | Output zones: 200 | Method: EQUAL AREA | Output: energy_slice
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - Con("energy_slice" == 200, 1)
 - Output: topsites
- Conversion Tools | From Raster | Raster to Polygon
 - Input: topsites | Output: topsites_poly
- Spatial Analyst Tools | Zonal | Zonal Statistics as Table
 - Zone data: topsites_poly | Zone field: OBJECTID | Value raster: energy_suitable
 - Statistic: MEAN | Output: sites_ranked
- Analysis Tools | Extract | Select
 - Input: topsites_poly | Output: top3_sites
 - Expression: OBJECTID = 9 OR OBJECTID = 4 OR OBJECTID = 3

6. Viewshed Analysis

- Data Management Tools | Features | Feature To Line
 - Input: top3_sites | Output: top3_lines
- Spatial Analyst Tools | Surface | Visibility
 - Input raster: dem.tif | Observer features: top3_lines | Output: viewshed
 - Observer parameters → Outer Radius: 20000
 - Environments → Cell size: 30 m
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - Con("viewshed" > 0, 1, 0)
 - Output: view_binary

7. Visibility per LSOA and Population Linkage

- Spatial Analyst Tools | Zonal | Zonal Statistics as Table
 - Zone data: Isoa.shp | Zone field: code | Value raster: view_binary
 - Statistic: ALL | Output: vis_table
- Add Data: census2011.csv
- Right-click Isoa → Joins and Relates → Add Join
 - Input Join Field: code | Join Table: census2011.csv | Join Table Field: geography code
- Right-click Isoa → Joins and Relates → Add Join
 - Input Join Field: code | Join Table: vis_table | Join Table Field: ZONE_CODE
- Right-click Isoa → Data → Export Features | Output: Isoa_vis_pop
- Right-click Isoa_vis_pop → Data → Export Table | Save as .csv to project folder

8. Clipping Results to Study Area a62

- Spatial Analyst Tools | Extraction | Extract by Mask (repeat for all raster outputs)
 - Input raster: (e.g. energy_suitable) | Mask: a62.shp | Output: (e.g. energy_a62)
- Analysis Tools | Extract | Clip (repeat for all vector outputs)
 - Input: (e.g. top3_sites) | Clip Features: a62.shp | Output: (e.g. top3_a62)

End of Appendix

